interchange means that their shapes and fundamental geometrical structures are quite different.5

Excited by the apparent contact we had made with the Dixon-Lerche-Vafa-Warner guess, Plesser and I pressed on to the linchpin question: Beyond the number of families of particles, do the two different Calabi-Yau spaces agree on the rest of their physical properties? After a couple more months of detailed and arduous mathematical analysis during which we received valuable inspiration and encouragement from Graham Ross, my thesis advisor at Oxford, and also from Vafa, Plesser and I were able to argue that the answer was, most definitely, yes. For mathematical reasons having to do with the even-odd interchange, Plesser and I coined the term mirror manifolds to describe the physically equivalent yet geometrically distinct Calabi-Yau spaces.6 The individual spaces in a mirror pair of Calabi-Yau spaces are not literally mirror images of one another, in the sense of everyday usage. But even though they have different geometrical properties, they give rise to one and the same physical universe when used for the extra dimensions in string theory.

The weeks after finding this result were an extremely anxious time. Plesser and I knew that we were sitting on an important new piece of string physics. We had shown that the tight association between geometry and physics originally set down by Einstein was substantially modified by string theory: Drastically different geometrical shapes that would imply different physical properties in general relativity were giving rise to identical physics in string theory. But what if we had made a mistake? What if their physical implications did differ in some subtle way that we had missed? When we showed our results to Yau, for example, he politely but firmly claimed that we must have made an error; he asserted that from a mathematical standpoint our results were far too outlandish to be true. His assessment gave us substantial pause. It’s one thing to make a mistake in a small or modest claim that attracts little attention. Our result, though, was suggesting an unexpected step in a new direction that would certainly engender a strong response. If it were wrong, everyone would know.

Finally, after much checking and rechecking, our confidence grew and we sent our paper off for publication. A few days later, I was